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FACULTY OF ELECTRICAL AND ELECTRONICS ENGINEERING
DEPARTMENT OF ELECTRONICS ENGINEERING

CAPSTONE PROJECT 2

DESIGN AND IMPLEMENTATION OF A PIPELINED RISC-V PROCESSOR INTEGRATING TILELINK-UL BUS PROTOCOL

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1. Introduction



Motivation

- Legacy processor core with **rigid dependency** on custom memory connections
- Significant limitations in system **modularity** and scalability
- Obstacles in integrating additional **peripheral devices**
- Necessity of adopting industry-standard **bus protocols**
- Transitioning towards a robust **System-on-Chip (SoC)** foundation

1. Introduction



Project Objectives

- Enable **TileLink-UL** communication within the RISC-V processor
- Implement standardized **Host and Device Adapters**
- Handle **misaligned memory accesses** via specialized logic
- Execute rigorous **functional verification** through simulation
- Validate hardware feasibility on **Intel Cyclone V FPGA**

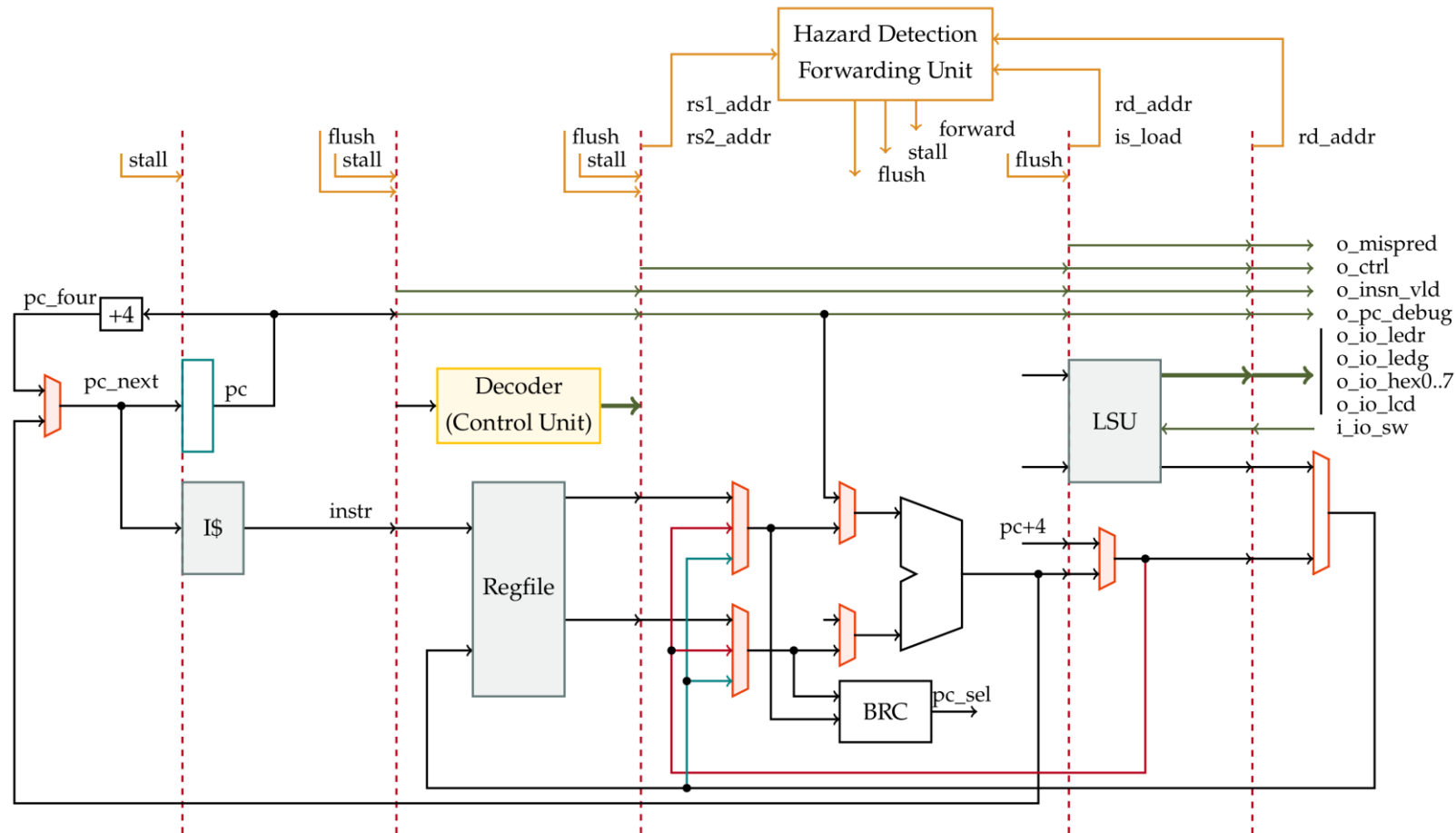
2. Theoretical Background



RISC-V Architecture

- Base integer instruction set: **RV32I**
- Classic **5-stage pipeline** microarchitecture
- Standard stages: IF, ID, EX, **MEM**, and WB
- Load-Store Unit (LSU) timing and signal requirements

2. Theoretical Background



Datapath of 5-Stage RV32I Pipelined Processor

2. Theoretical Background



TileLink-UL Protocol

- **Master-slave** communication topology
- Independent **Channel A** (Request) and **Channel D** (Response)
- Decoupled **valid-ready** handshake mechanism
- Simplified operations: **Get** (Read) and **Put** (Write)

2. Theoretical Background



Channel	Signal Name	Direction	Description
Channel A	a_valid	Master to Slave	Indicates that the request information is valid.
	a_ready	Slave to Master	Indicates that the slave is ready to accept the request.
	a_opcode [2:0]	Master to Slave	Specifies the operation type (Get, PutFullData, PutPartialData).
	a_addr [AW-1:0]	Master to Slave	The target byte address for the transaction.
	a_data [DW-1:0]	Master to Slave	The data payload to be written (for Put operations).
	a_mask [DBW-1:0]	Master to Slave	Byte enable mask for writing specific bytes within a word.

Key Signals of the TileLink-UL Protocol - Channel A

2. Theoretical Background



Channel	Signal Name	Direction	Description
Channel D	d_valid	Slave to Master	Indicates that the response information is valid.
	d_ready	Master to Slave	Indicates that the master is ready to accept the response.
	d_opcode [2:0]	Slave to Master	Specifies the response type (AccessAck, AccessAckData).
	d_data [DW-1:0]	Slave to Master	The data payload read from memory (for Get operations).

Key Signals of the TileLink-UL Protocol - Channel D

2. Theoretical Background



Interleaved Memory

- Four independent **8-bit memory banks**
- Supports **misaligned accesses** across word boundaries
- Requirement for **address swizzling** translation
- Simultaneous access to different **row indices**
- Bridging flat-addressed bus with **interleaved banks**

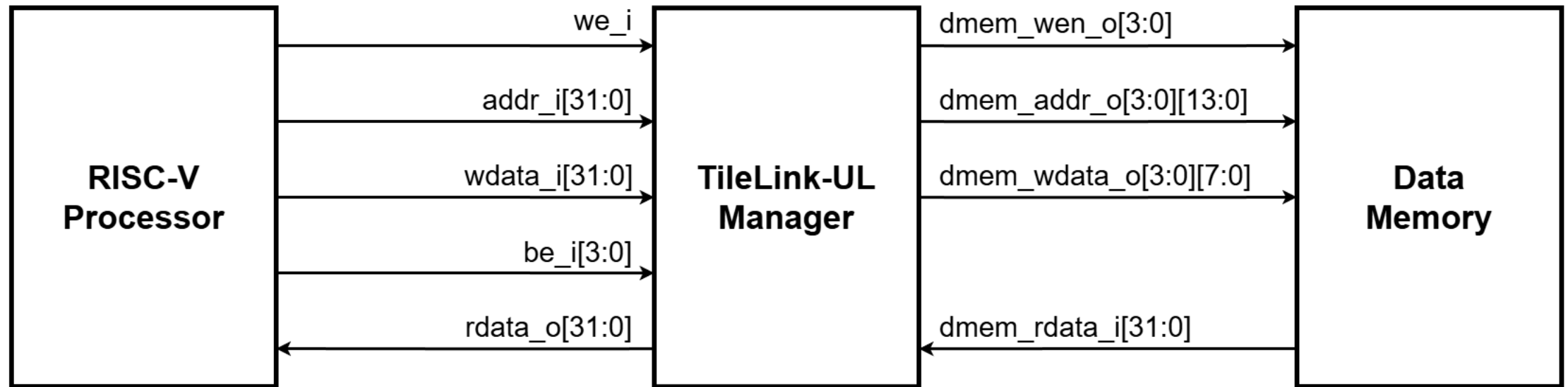
3. Design - Implementation



Top-Level Architecture

- Integration of RISC-V core and **Interconnect Manager**
- Standardized **SoC abstraction** layer
- System clock source: **50 MHz**

3. Design - Implementation



Top-Level System Architecture

3. Design - Implementation



TileLink Manager

- Enforcement of **standard interfaces** between CPU and memory
- Direct mapping of **32-bit addresses** to Host Adapter
- Byte-granularity support via **write masks**

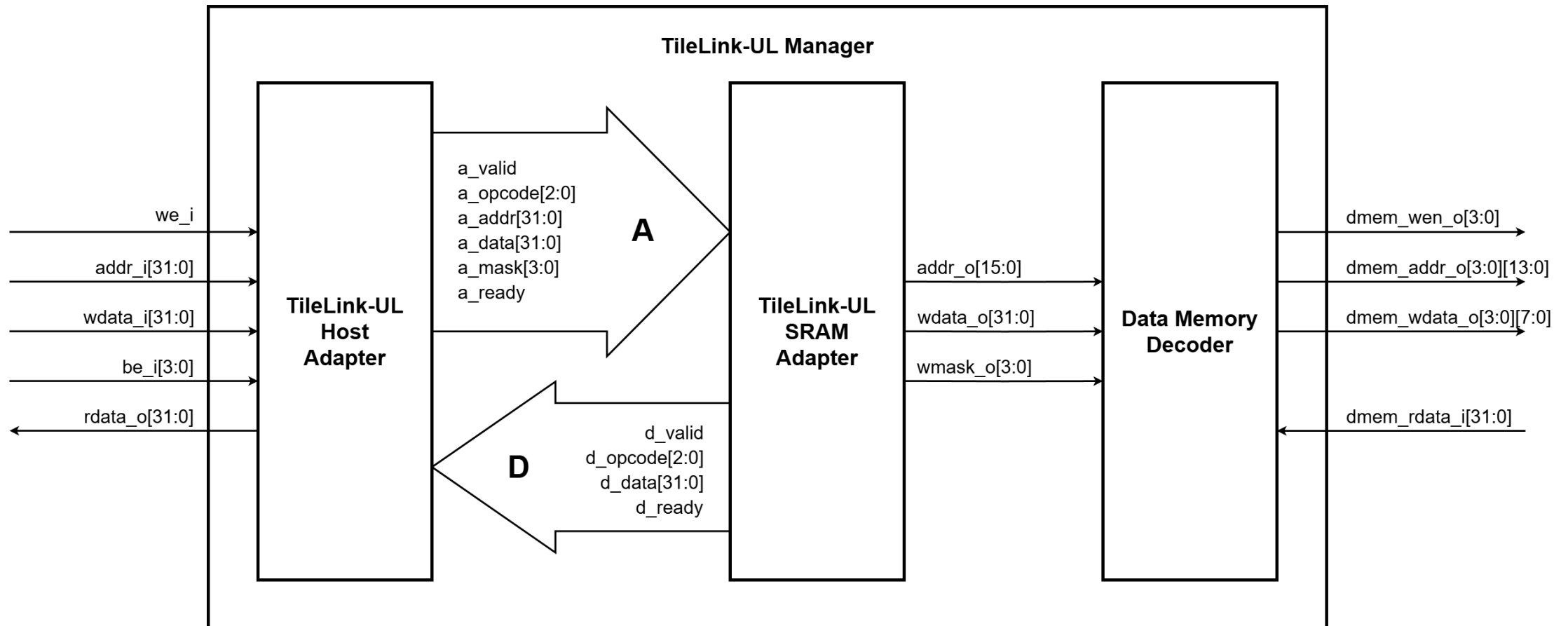
3. Design - Implementation



TileLink Manager

- Enforcement of **standard interfaces** between CPU and memory
- Direct mapping of **32-bit addresses** to Host Adapter
- Byte-granularity support via **write masks**

3. Design - Implementation



Internal Structure of TileLink Manager

3. Design - Implementation



Interface	Signal Name	Width	Direction	Description
System	clk	1-bit	Input	System clock signal.
	rstn	1-bit	Input	Active-low asynchronous reset.

TileLink Manager Signals - System

3. Design - Implementation



Interface	Signal Name	Width	Direction	Description
Processor Side	req_i	1-bit	Input	Memory access request strobe (from LSU).
	we_i	1-bit	Input	Write Enable control (1 = Write, 0 = Read).
	addr_i	32-bit	Input	Target byte address (from ALU Result).

TileLink Manager Signals – Processor Side

3. Design - Implementation



Interface	Signal Name	Width	Direction	Description
	wdata_i	32-bit	Input	Data to be written to memory.
	be_i	4-bit	Input	Byte-enable mask (Active high).
	rdata_o	32-bit	Output	Data read from memory returned to CPU.

TileLink Manager Signals – Processor Side

3. Design - Implementation



Interface	Signal Name	Width	Direction	Description
Physical Memory Side	dmem_addr_o	4 x 14-bit	Output	Independent physical addresses for 4 memory banks (Decoded).
	dmem_wdata_o	4 x 8-bit	Output	Write data split for 4 memory banks.
	dmem_wen_o	4-bit	Output	Independent Write Enable for each bank.
	dmem_rdata_i	32-bit	Output	Combined read data from 4 memory banks.

TileLink Manager Signals – Memory Side

3. Design - Implementation



Interface	Signal Name	Width	Direction	Description
TL-UL Channel A (Request)	tl_o.a_valid	1-bit	Output	Master Valid: Request is valid.
	tl_o.a_opcode	3-bit	Output	Operation type (Get, PutFull, PutPartial).
	tl_o.a_address	32-bit	Output	Target address on the bus.
	tl_o.a_mask	4-bit	Output	Byte lane mask for the bus.
	tl_o.a_data	3-bit	Output	Write data payload on the bus.
	tl_i.a_ready	1-bit	Input	Slave Ready: Slave accepts the request.

TileLink-UL Bus – Channel A

3. Design - Implementation



Interface	Signal Name	Width	Direction	Description
TL-UL Channel D (Response)	tl_i.d_valid	1-bit	Input	Slave Valid: Response is available.
	tl_i.d_opcode	3-bit	Input	Response type (AccessAck, AccessAckData).
	tl_i.d_data	32-bit	Input	Read data payload from the bus.
	tl_o.d_ready	1-bit	Output	Master Ready: Host accepts response.

TileLink-UL Bus – Channel D

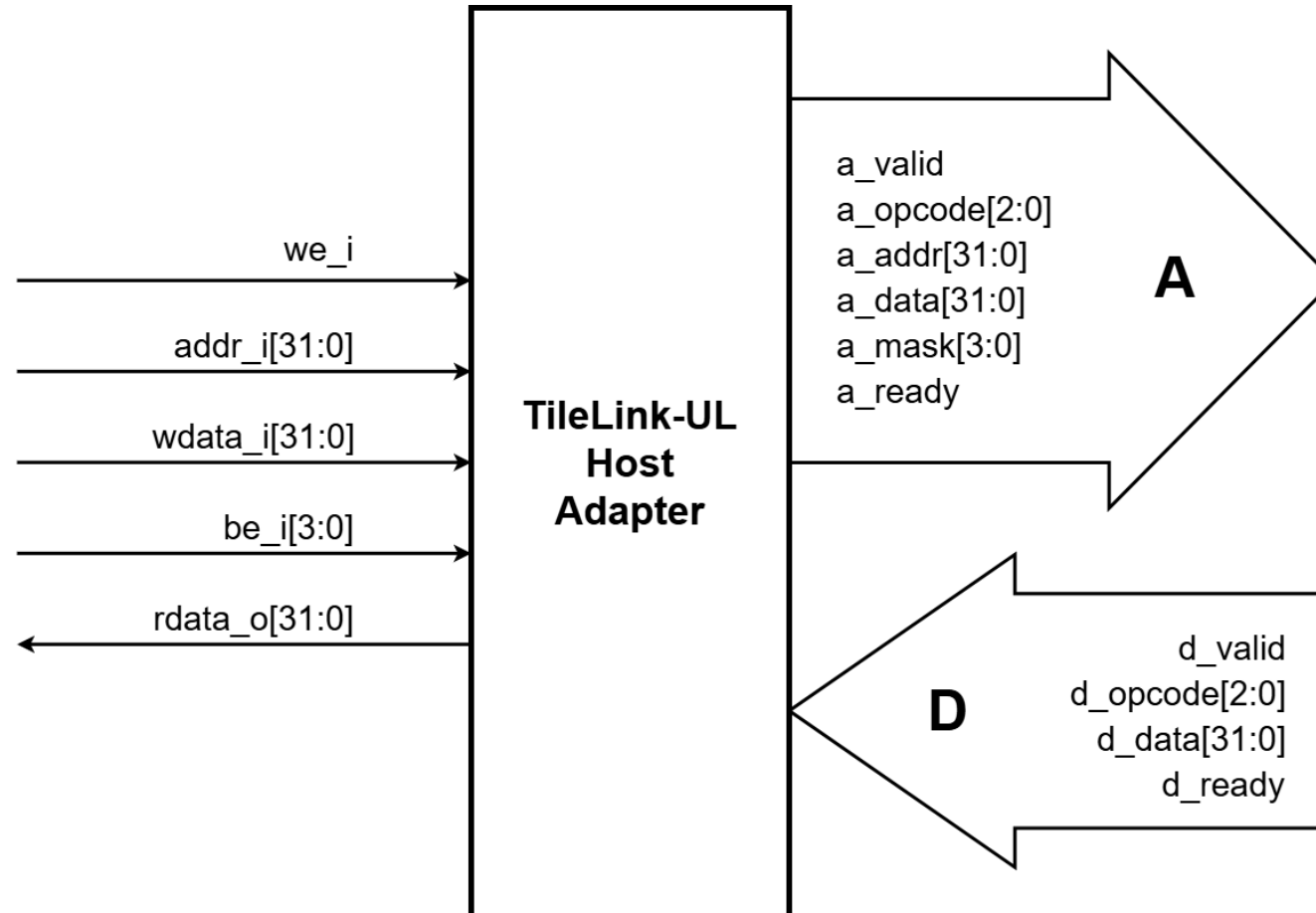
3. Design - Implementation



Host Adapter

- **Opcode generation** logic (Get, PutFull, PutPartial)
- Request phase control via **a_valid** and **a_ready**
- Response monitoring for **d_valid** and **data capture**

3. Design - Implementation



Host Adapter Logic Flow

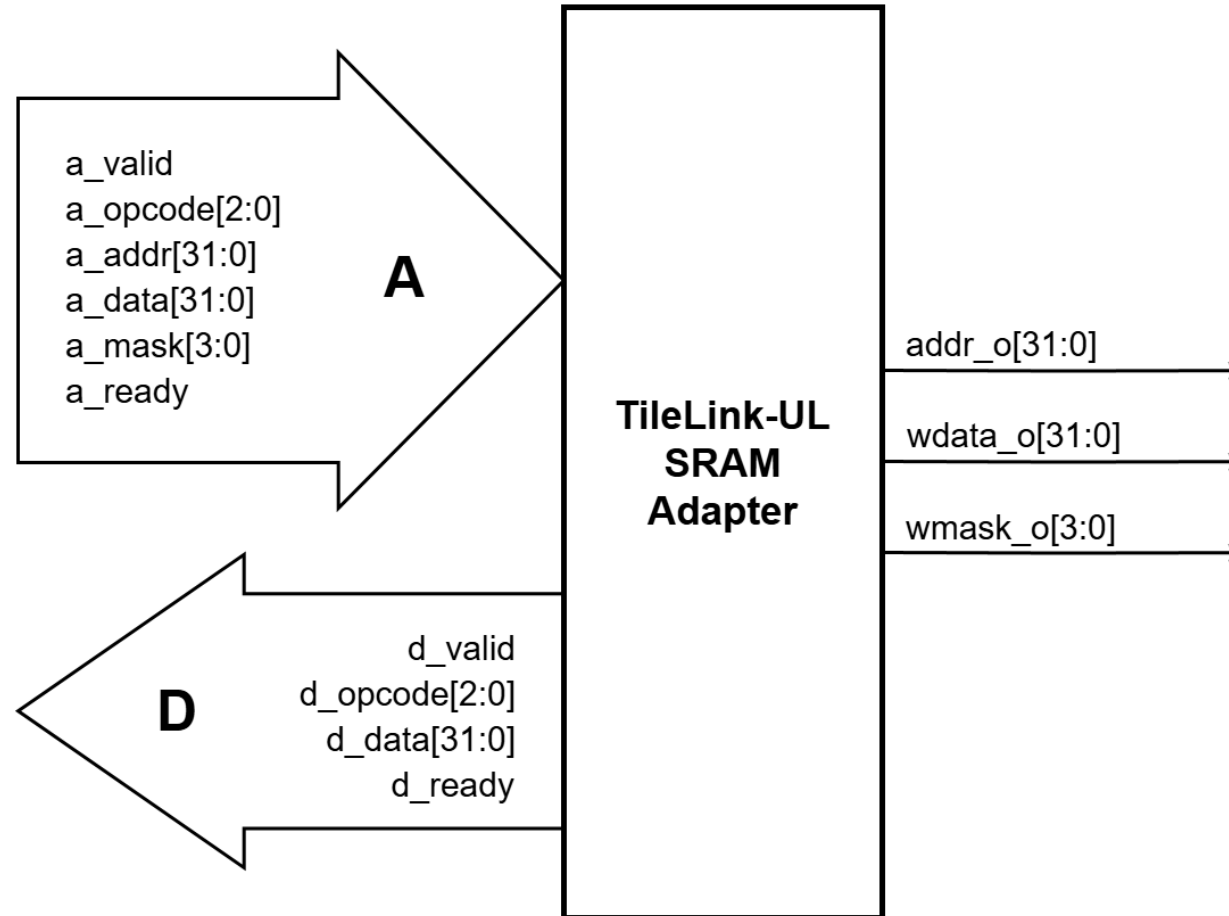
3. Design - Implementation



Device Adapter

- Automatic **address truncation** for 64 KB memory space
- Command decoding for **SRAM control** lines
- **Synchronous acknowledgment** loop for 1-cycle latency

3. Design - Implementation



Device Adapter Logic Flow

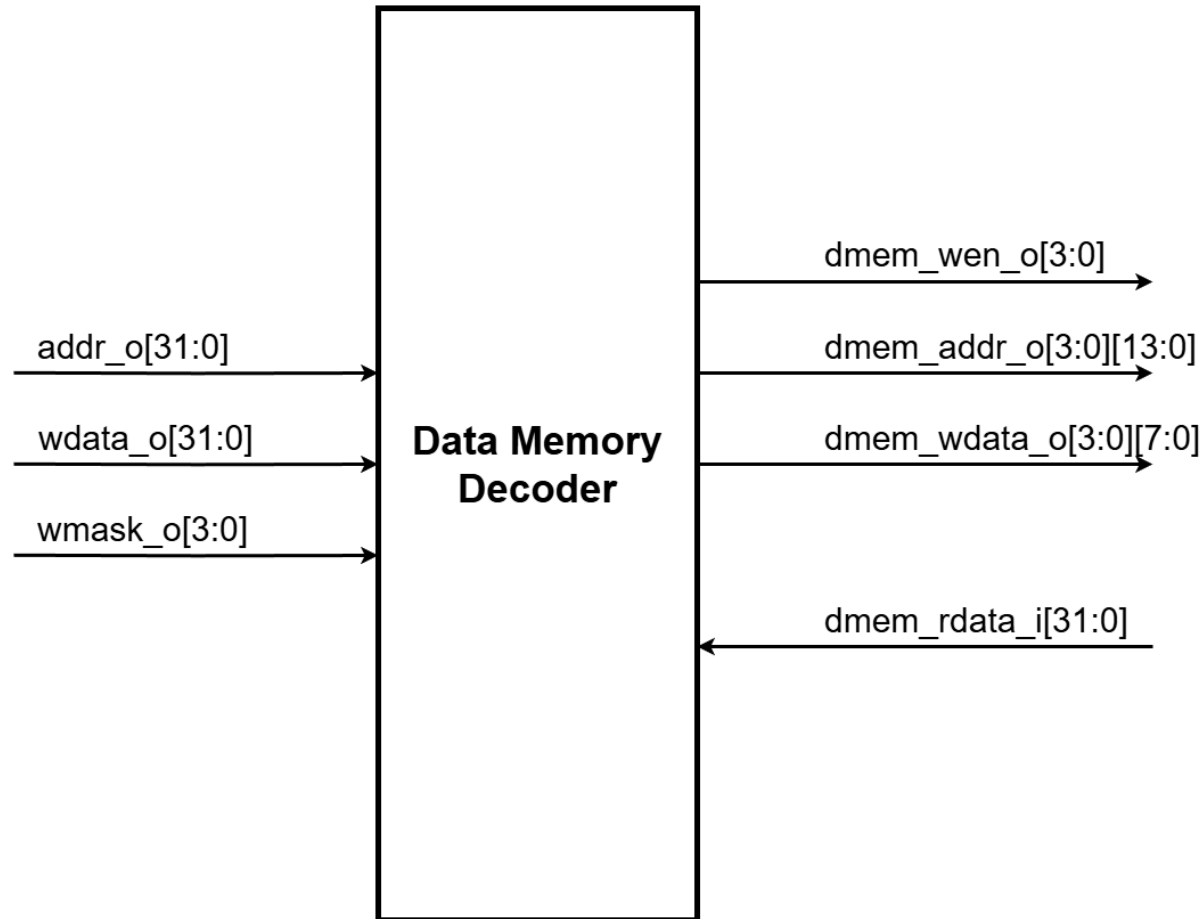
3. Design - Implementation



Address Decoupling Logic

- **Misalignment handling** for Offset = 0, 1, 2, 3
- Dynamic calculation of **idx_base** and **idx_next**
- **Bank-specific** write enable gating logic

3. Design - Implementation



Address Decoupling Circuit

3. Design - Implementation



Signal Name	Width	Direction	Description
sram_addr_i	16-bit	Input	Base address received from Device Adapter.
sram_wdata_i	32-bit	Input	Full 32-bit write data payload.
sram_wmask_i	4-bit	Input	Write mask (strobe) from Device Adapter.
dmem_addr_o	4 x 14-bit	Output	Independent row index calculated for each of the 4 banks.
dmem_wdata_o	4 x 8-bit	Output	Write data sliced into 8-bit segments for each bank.
dmem_wen_o	4-bit	Output	Independent write enable for each bank (gated by logic).

Address Decoupling Circuit

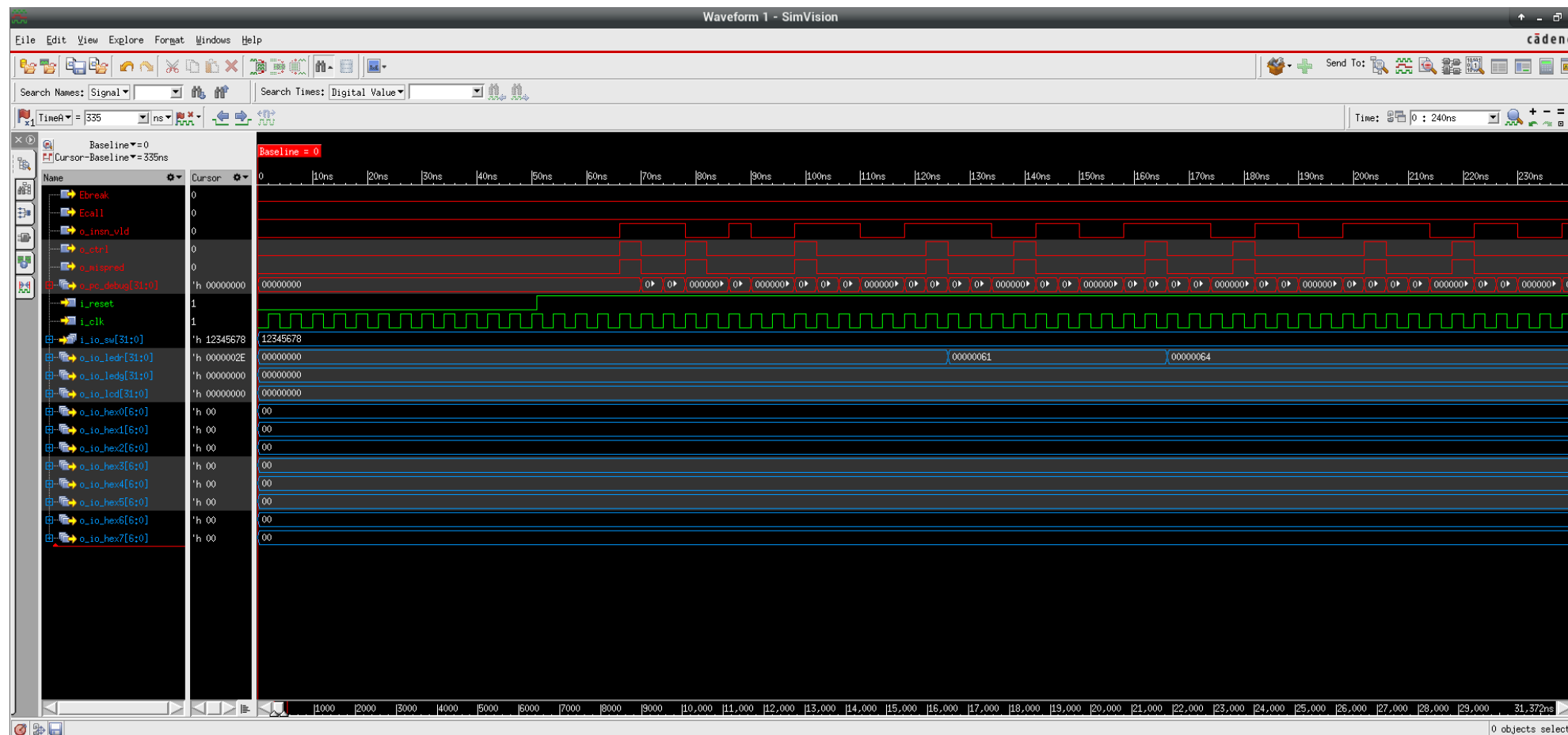
4. Implementation Results



Functional Verification

- Visual timing verification using **SimVision waveforms**
- Full **RV32I compliance** via automated test suites
- Successful verification of **misaligned accesses**
- Achieved Instruction Per Cycle (**IPC**) ratio of **0.62**

4. Implementation Results



SimVision Waveform demonstration

4. Implementation Results



SimVision Waveform demonstration (cont.)

4. Implementation Results



```
Design hierarchy summary:
      Instances  Unique
Modules:          193    52
Verilog packages:   3     6
Primitives:         5     1
Registers:         788   522
Scalar wires:      1179    -
Expanded wires:     410    17
Vectored wires:     590    -
Always blocks:      38     28
Initial blocks:     13     10
Cont. assignments: 391    141
Pseudo assignments: 827   464
Compilation units:   1      1
Simulation timescale: 1ps

Writing initial simulation snapshot: worklib.mux_8:sv
Loading snapshot worklib.mux_8:sv ..... Done
xmsim: *W,DSEM2009: This SystemVerilog design is simulated as per IEEE 1800-2009 SystemVerilog simulation semantics. Use -disable_sem2009 option for turning off SV 2009 simulation semantics.
xcelium> source /opt/cadence/XCELIUM2009/tools/xcelium/files/xmsimrc
xcelium> run

PIPELINE - ISA tests

add.....PASS
addi.....PASS
sub.....PASS
and.....PASS
andi.....PASS
or.....PASS
ori.....PASS
xor.....PASS
xori.....PASS
slt.....PASS
slti.....PASS
sltu.....PASS
sltiu.....PASS
sll.....PASS
```

Xcelium Simulation Log

4. Implementation Results



```
sll.....PASS
slli.....PASS
srl.....PASS
srli.....PASS
sra.....PASS
srai.....PASS
lw.....PASS
lh.....PASS
lhu.....PASS
lb.....PASS
lbu.....PASS
sw.....PASS
sh.....PASS
sb.....PASS
auipc.....PASS
lui.....PASS
beq.....PASS
bne.....PASS
blt.....PASS
bltu.....PASS
bge.....PASS
bgeu.....PASS
jal.....PASS
jalr.....PASS
malign.....PASS
iosw.....PASS

===== Result =====
Total Clock Cycles Executed = 7830
Total Instructions Executed = 4825
Total Branch Instructions   = 1450
Total Branch Mispredictions = 1450

-----
Instruction Per Cycle (IPC) = 0.62
Branch Misprediction Rate  = 100.00 %

END of ISA tests
```

Xcelium Simulation (cont.)

4. Implementation Results



Synthesis & Resources

- Target Device: **Intel Cyclone V 5CSXFC6D6F31C6**
- Low logic overhead: **5% ALM** utilization (1,990 ALMs)
- Memory usage: **9% Block RAM** bits (526,336 bits)
- Operating frequency: **60.16 MHz Fmax**
- Positive timing slack: **3.4 ns** safety margin

4. Implementation Results



The screenshot displays the Quartus Prime Lite Edition interface. The main window shows the 'Flow Summary' report, which details the compilation process and resource utilization. The 'Messages' window at the bottom shows the output of the compilation, including warnings and errors.

Flow Summary

Flow Status	Successful - Wed Dec 31 04:58:39 2025
Quartus Prime Version	18.1.0 Build 625 09/12/2018 S.J. Lite Edition
Revision Name	top
Top-level Entity Name	top
Family	Cyclone V
Device	5CSXFC6D6F31C6
Timing Models	Final
Logic utilization (in ALMs)	1,990 / 41,910 (5 %)
Total registers	910
Total pins	225 / 499 (45 %)
Total virtual pins	0
Total block memory bits	526,336 / 5,662,720 (9 %)
Total DSP Blocks	0 / 112 (0 %)
Total HSSI RX PCSs	0 / 9 (0 %)
Total HSSI PMA RX Deserializers	0 / 9 (0 %)
Total HSSI TX PCSs	0 / 9 (0 %)
Total HSSI PMA TX Serializers	0 / 9 (0 %)
Total PLLs	0 / 15 (0 %)
Total DLLs	0 / 4 (0 %)

Messages

```
INFO: Analyzing Fast 1100mV OC Model
332123 Deriving Clock Uncertainty. Please refer to report_sdc in the Timing Analyzer to see clock uncertainties.
332148 Timing requirements not met
332146 Worst-case setup slack is -8.260
332146 Worst-case hold slack is 0.176
332140 No Recovery paths to report
332140 No Removal paths to report
332146 Worst-case minimum pulse width slack is -2.636
332102 Design is not fully constrained for setup requirements
332102 Design is not fully constrained for hold requirements
Quartus Prime Timing Analyzer was successful. 0 errors, 6 warnings
293000 Quartus Prime Full Compilation was successful. 0 errors, 479 warnings
```

Logic Resource Utilization Report

4. Implementation Results



The screenshot displays the Quartus Prime Lite Edition interface. The main window shows the 'Flow Summary' for a project named 'top'. The summary indicates a successful compilation on Wednesday, December 31, 2025, at 04:58:39. The device used is Cyclone V 5C5XFC6D6F31C6. Key resource utilization metrics are listed:

Resource	Used	Available	Usage (%)
Total block memory bits	526,336	5,662,720	9 %
Total DSP Blocks	0	112	0 %
Total HSSI RX PCSs	0	9	0 %
Total HSSI PMA RX Deserializers	0	9	0 %
Total HSSI TX PCSs	0	9	0 %
Total HSSI PMA TX Serializers	0	9	0 %
Total PLLs	0	15	0 %
Total DLLs	0	4	0 %

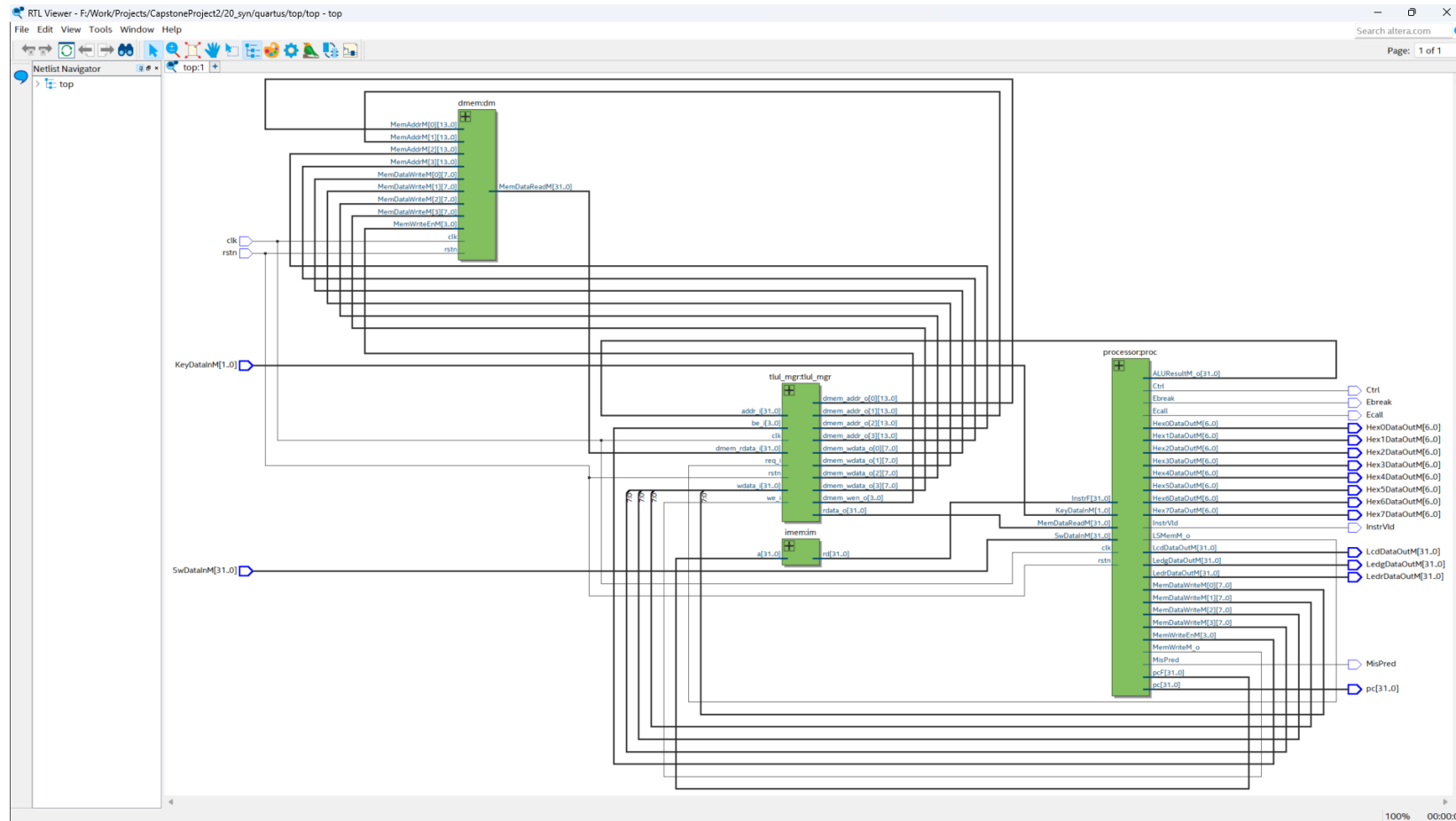
The 'Messages' window at the bottom shows the following output:

```
INFO: Analyzing Fast 1100mV OC Model
332123 Deriving Clock Uncertainty. Please refer to report_sdc in the Timing Analyzer to see clock uncertainties.
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Quartus Prime Timing Analyzer was successful. 0 errors, 6 warnings
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```

Total Block Memory Bits Report



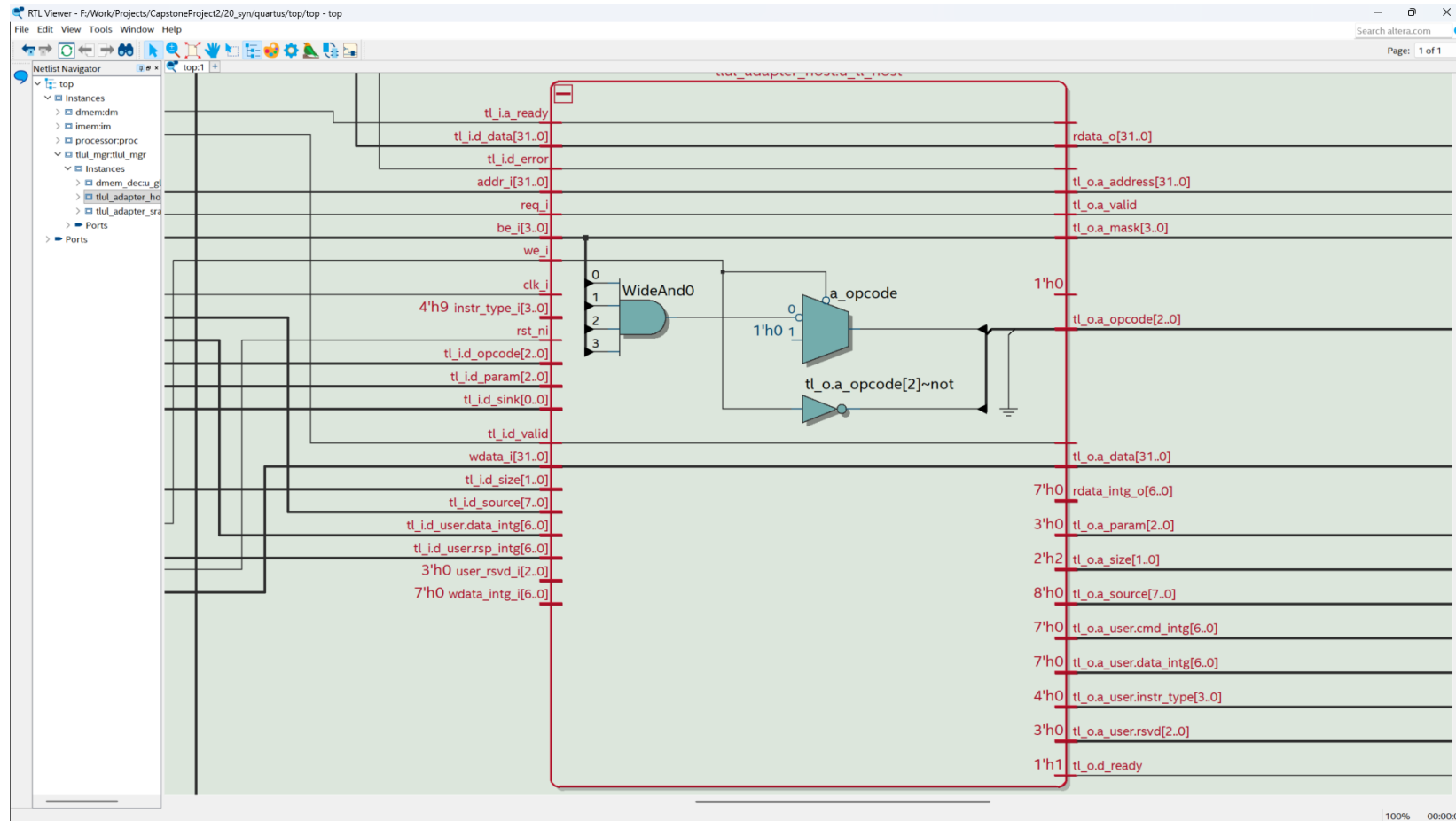
4. Implementation Results



Netlist of the interconnection between the RISC-V Core and the TileLink Manager

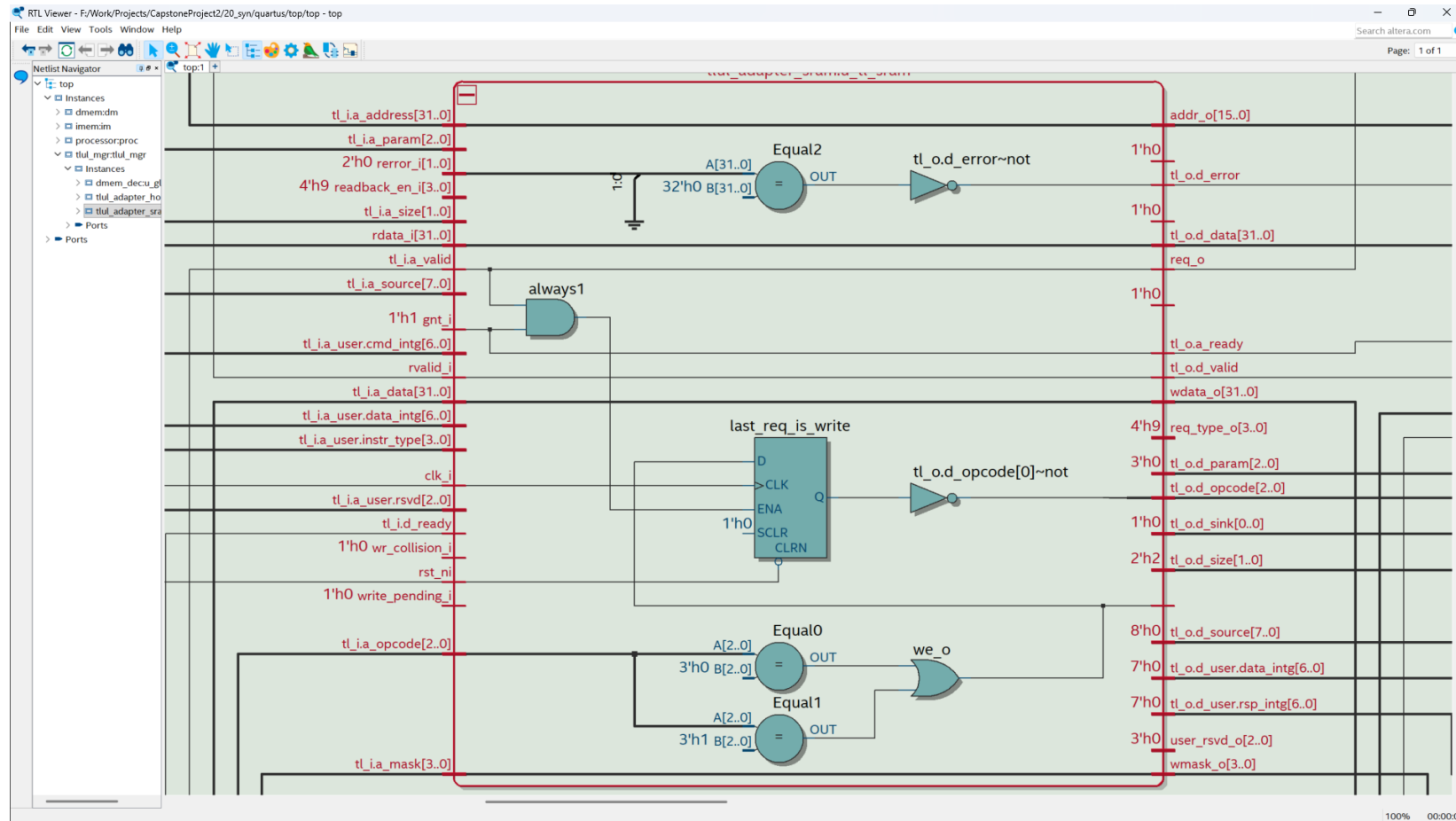


4. Implementation Results



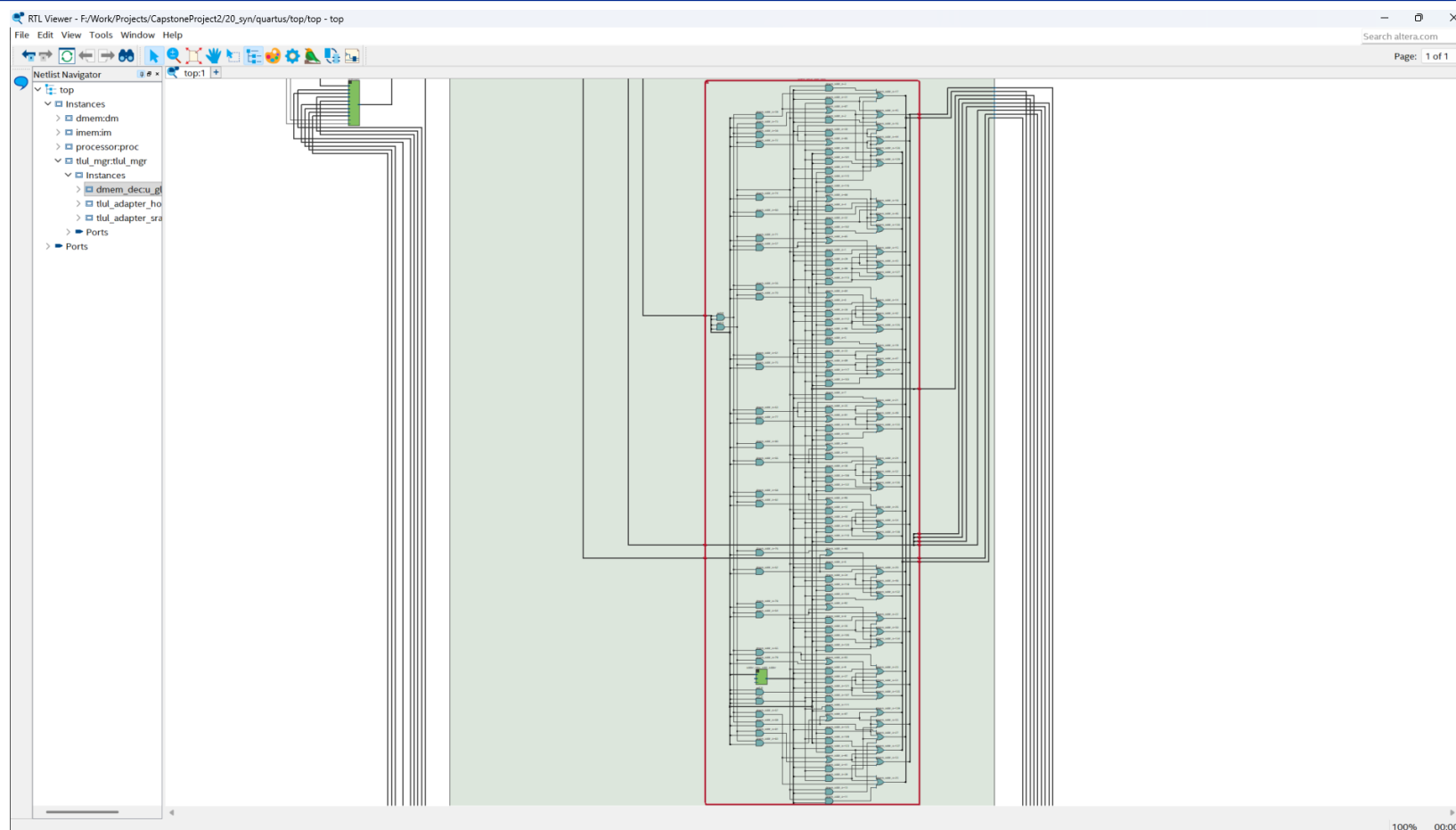
Netlist of the Host Adapter Module

4. Implementation Results



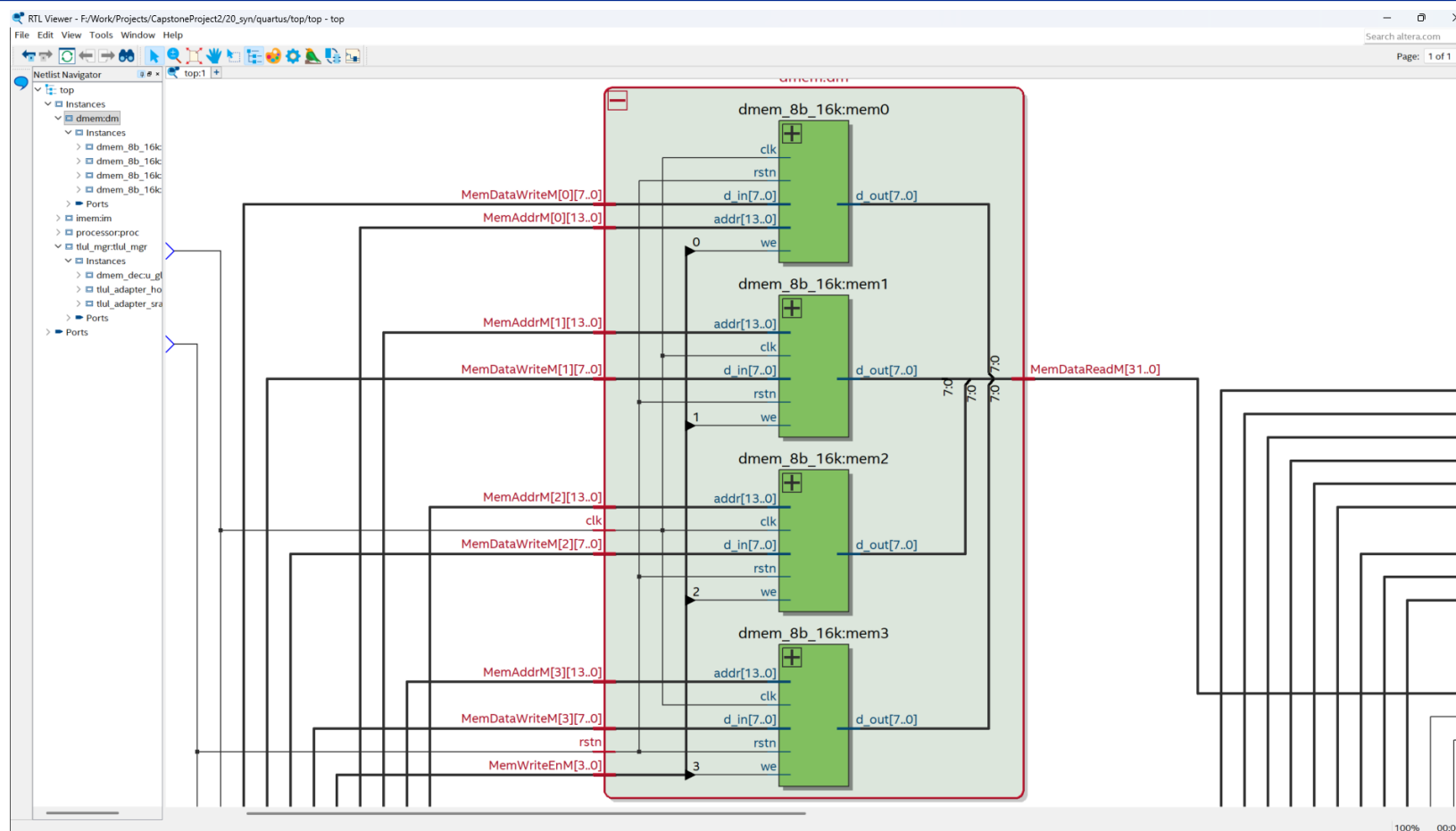
Netlist of the Device Adapter Module

4. Implementation Results



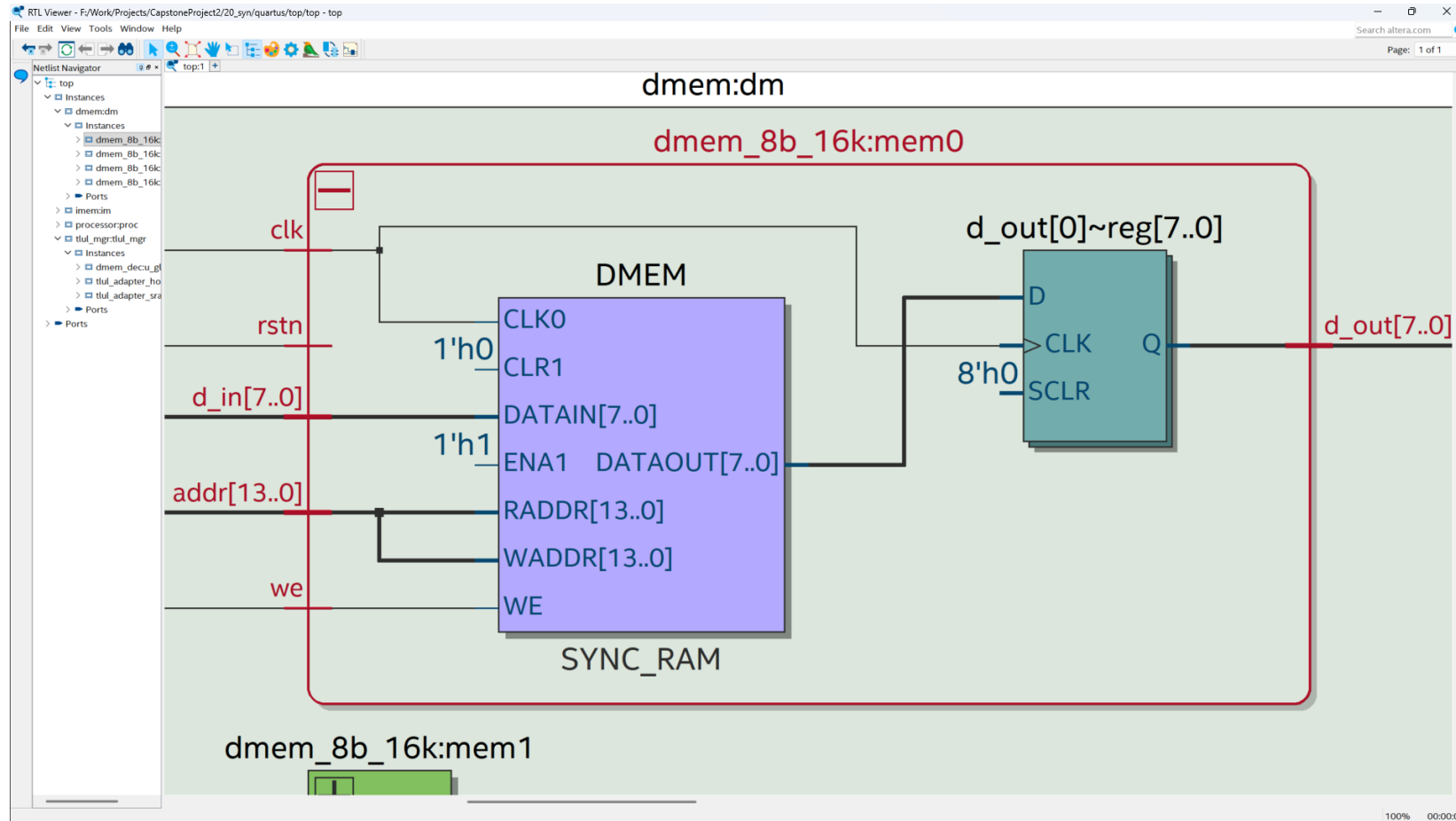
Netlist of the Address Decoupling Logic

4. Implementation Results



Netlist of the Data Memory (64Kb)

4. Implementation Results



Netlist of a 8-bit Memory Bank (16 Kb)

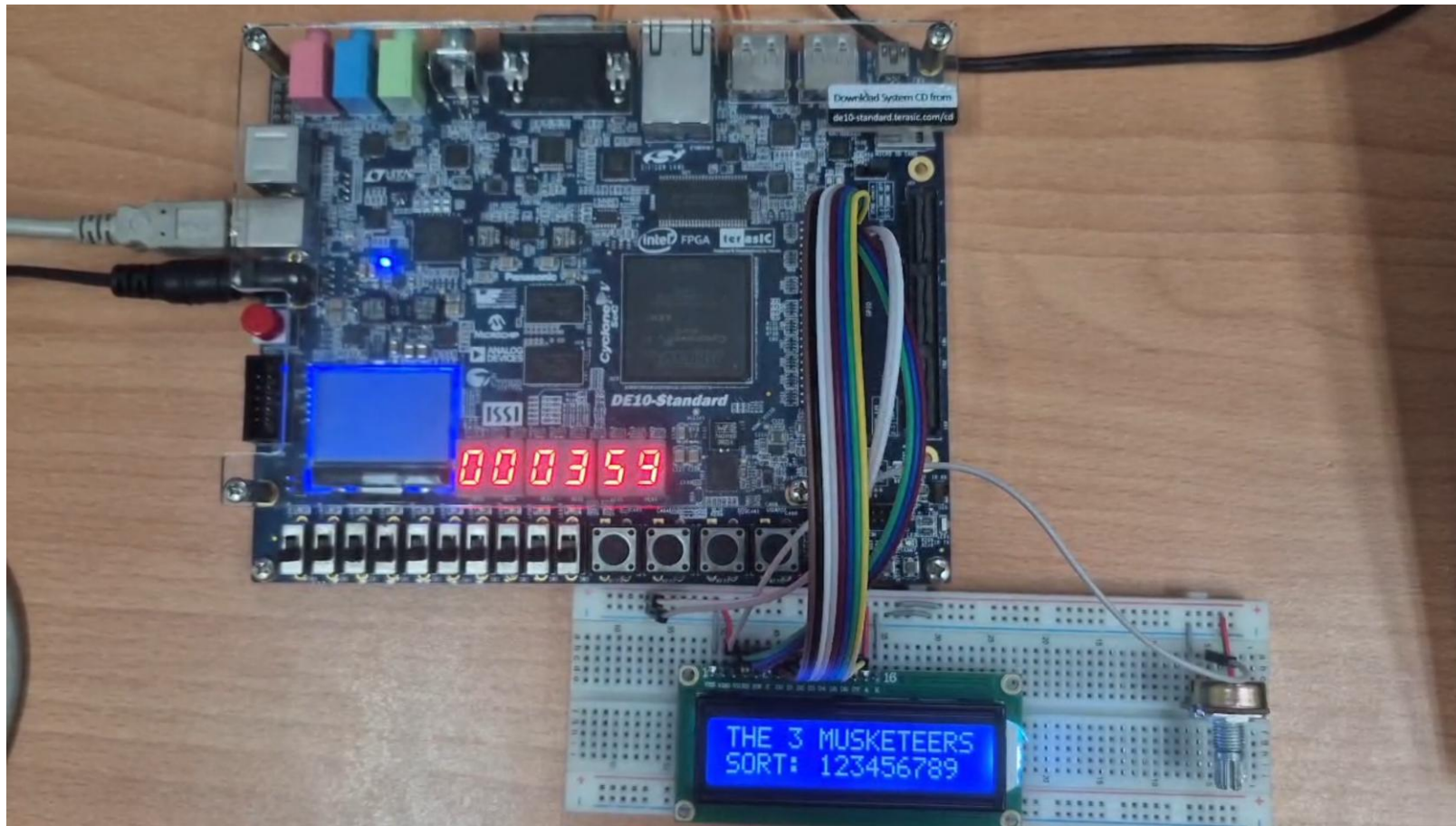
4. Implementation Results



Hardware Implementation

- Successful deployment on **DE10-Standard** kit
- Functional **Stopwatch** demo application
- Real-time data display on **LCD**
- Validated **MMIO** routing through TileLink adapters

4. Implementation Results



Experimental setup utilizing the DE10-Standard development board

5. Conclusion - Future Vision



Conclusion

- Achieved core-to-memory **decoupling** via TileLink-UL
- Robust handling of **misaligned accesses** verified
- Successful **FPGA validation** with minimal resource overhead
- Compliant with **industry standards** for modular SoC
- Established a **scalable framework** for future expansion

5. Conclusion - Future Vision



Future Vision

- Implementation of a 1:N Bus Crossbar interconnect
- Integration of Level-1 Instruction Cache
- Support for Direct Memory Access (DMA) controllers
- Exploration of TileLink-UH for burst transactions
- Integration of communication peripherals (UART, SPI, I2C)



**THANK YOU
FOR YOUR ATTENTION.**